Remote Controlled Plane and Controller

ENGR 2999, Douglas College

Maximus Larouche, Brendan Morrison

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Abstract

This capstone project details the development of a semi-autonomous remote-controlled aircraft and its custom handheld controller. The project transitioned from an initial 3D-printed design to a 1250g laser-cut plywood airframe to ensure structural rigidity for a high-power 4S propulsion system. Aerodynamic analysis utilizing the lift equation resulted in a 62.5-inch wingspan, providing a 55% lift surplus and a significant safety margin for flight testing. The integrated electronics system features a custom RP2350-based control board and a 2.4GHz LoRa radio link, achieving a 1.5:1 thrust-to-weight ratio that places the aircraft in the "sport flyer" category. While core flight systems, including 100Hz PWM signal transmission and tail-dragger landing gear, are fully operational, secondary features such as IMU-based stability and OLED telemetry were deferred to prioritize link reliability. This platform serves as a functional foundation for future multicore software optimizations and real-world environmental validation.

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Introduction

For our project, the moto for our plane is as follows: "*it only has to fly, but that doesn't mean it has to land*". The plane was originally based around the Cessna 182 from which we steered our own design to fit our needs and constraints. The plane's wings, aerodynamics, and power systems were originally designed for around a 1:1 thrust to weight ratio, however the final weight was around 300g less than our original target, thus resulting in a 1.5:1 ratio, putting it comfortably in the sport flyer category.

The plane uses a 9x6 2-blade propeller with 1450kV brushless motor, powered by a 60A ESC and a 2200mAh 4S LiPo battery. The plane's communication comes from a 2.4GHz LoRa radio antenna, with a custom soldered RP2350 based control board handling the communication and servo control. The controller uses a slide potentiometer for the throttle and a joystick for the pitch and roll. The controller uses the same MCU and antenna, with a built-in battery and LiPo charging circuit feeding the board.

The software, while functional, is still undergoing improvements and optimizations to allow greater levels of control, and stability. The software is written in Arduino with the ArduinoPico core, with the current version only operating on the single core of the RP2350 on both devices. The software is written to be lightweight and efficient to allow for the most resources to be dedicated to transmitting and receiving data for the antenna.

On the mechanical side the project was initially conceived as a 3D-printed project, the development underwent a significant pivot toward 2D CAD and laser-cut plywood construction to meet fabrication deadlines and ensure a robust platform for the high power propulsion system. The mechanical team focused on optimizing wing area via aerodynamic lift equations and implementing a tail-dragger landing gear configuration to maximize ground clearance. This platform serves as the foundation for the integrated avionics and flight control software.

Bill of Materials

Item	**Quantity**	**Unit Cost**	**Cost**
Soldering / Prototyping Materials	1 (pack)	$16.99	$16.99
1450Kv Motor	1	$30.81	$30.81
60A ESC	1	$44.79	$44.79
14.8v 2200mAh Battery	1	$32.99	$32.99
IMU/Gyroscope module (MPU6050)	1	$4.33	$4.33
Slide Potentiometers	1	$2.55	$2.55
Toggle Switches	1	$1.07	$1.07
NRF24L01 PA + LNA Antenna Module	2	$4.95	$9.90

WaveShare RP2350-Zero modules	2	$11.50	$22.99
Joystick Modules	1	$5.75	$5.75
MG90s 9g Servos	3	~$4.43	~$13.30
9x6 Two Blade Propeller	1	$8.90	$8.90
Miscellaneous electrical components (already had / found them)	○ The Insides of a CAT6 Ethernet Cable (wire) ○ SSD1306 128x84 OLED Display (1x) ○ EC11 Rotary Encoder (1x) ○ XL6009 Boost Converter (1x) ○ TP4056 Battery Charger (1x) ○ Scavenged Lithium Ion Cell (1x) ○ 10k Resistors (5x) ○ 100uF Electrolytic Capacitor (2x) ○ 10nF Ceramic Capacitors (7x)		
Landing Gear	1	$21.99	$21.99
Packing Tape	1	$2.99 (ON SALE)	$2.99
Foam Board	3	$1.25	$3.75
Control Surface Hardware	1	$18.99	$18.99
¼" Wooden Dowel	1	$3.97	$7.94
Miscellaneous mechanical components (already had / found them)	○ ~500g Plywood ○ M3 Bolts, Nuts, Lock Washers (x6) ○ M4 Bolts, Nuts (x2)		
		Total Cost:	$250.03

Design and Changelog

Mechanical Systems

A. Specifications:

1. Wing Loading and Lift: For a 62.5-inch wingspan with a 7-inch chord, the total wing area is approximately 437.5 square inches or 0.2823 square meters. At a total weight of ~1250g, the wing

loading is low enough to allow for stable flight at speeds achievable by a 1450KV motor and a 9-inch propeller.

2. Structural Material: The fuselage is constructed from 5mm laser-cut plywood using friction fit construction with a small caveat. This provides the necessary torsional rigidity to support the battery and motor torque that wouldn't be possible using only foam in the fuselage. The small caveat is the friction fit in the middle of the plane started to fail upon final assembly and is now aided with tape and epoxy along the area that joins together.
3. Landing Gear Configuration: A tail-dragger configuration was selected to maximize the propeller clearance (~5 inches) and simplify assembly due to time constraints. This configuration requires the main gear to be positioned 3 inches ahead of the Centre of Gravity to prevent "nosing over" during taxi.
4. Propeller Clearance: For a 9-inch propeller, a minimum ground clearance of 0.5 inches at a level position is required. In the three-point stance, the tail-dragger setup further increases this clearance to ~2.5 inches, protecting the drivetrain during takeoff.
5. Control Surface Proportion: The elevator and ailerons were designed to be approximately 30% of the wing and horizontal stabilizer chord respectively. This ratio is considered good practice in RC Plane design to ensure sufficient pitch authority for a short-couple 28-inch (0.700m) fuselage without overwhelming the 9g servos.

B. Math:

1. A primary objective of the design phase was determining the optimal wing area (S) required to generate sufficient lift for the airframe. The design was calculated based on an initial target mass of 1.5 kg to provide a conservative safety margin. In the absence of wind tunnel data for the specific Armin wing profile, established aerodynamic constants and conservative RC aircraft coefficients were utilized in the lift equation. This "worst-case" design approach ensured that the resulting 62.5-inch wingspan would possess a substantial lift surplus, accommodating the final system weight while maintaining a low stall velocity for safe flight testing.

$$S = Minimum\ Wing\ Area$$

$$L = Lift = (1.50\ kg)\left(9.81\frac{m}{s^2}\right) = 12.3\ N$$

$$\rho = Air\ Density = 1.225\frac{kg}{m^3}$$

$$v = Target\ Takeoff\ Speed = 10\ m/s$$

$$C_L = Max\ Coefficient\ of\ Lift = 1.1$$

$$L = \frac{1}{2}\rho v^2 S C_L$$

$$S = \frac{2L}{\rho v^2 C_L} = \frac{2(12.3)}{(1.225)(10)^2(1.1)} = 0.182\ m^2\ \approx 282\ in^2$$

Before construction, the minimum wing area (S) was calculated using the lift equation, assuming a target takeoff velocity of 10 m/s and a Coefficient of Lift of 1.1. While the mathematical minimum area required to support the mass was, a design area of (a 62.5-inch span) was selected. This provides a 55% lift surplus, lowering the stall speed and providing a broader safety margin for manual flight testing. Furthermore, this dimension was strategically chosen to align with the standard stock lengths of the foam board material, optimizing the building process by eliminating the need for span-wise cuts and reducing potential points of structural weakness at the wingtips.

The following is the changelog on the mechanical side:

Part	Description	Change List	Conclusions
Fuselage Front	The front/motor mount section of the aircraft	RCPlaneSkeletonFront1 • First draft of the front bringing the 3d sketch to reality RCPlaneSkeletonFront2 • Added cross members to the area that connects to rest of fuselage to increase strength (includes improved geometry) RCPlaneSkeletonFront3 • Improved Geometry RCPlaneFront1 • Design switch from 3d modelling and printing	RCPlaneSkeleton(Front/Middle/Tail) Conclusions: • Entering the project with limited 3D modeling experience presented significant challenges, causing the initial design phase to exceed the projected timeline before a presentable model was achieved. • The failure of the initial 3D-printed fuselage acted as a critical turning point. Recognizing that complex 3D modeling was no longer feasible within the remaining timeframe, the design philosophy shifted to 2D CAD and laser cutting. This allowed for the utilization of familiar fabrication

		• to 2d CAD and laser cutting RCPlaneFront2 • Added bolt holes to mount the motor to the front	methods to maintain project momentum. RCPlane(Front/Top) Conclusions: • Transitioning to laser-cut plywood proved to be the correct strategic decision. Although this resulted in a more utilitarian and less aerodynamic aesthetic, it enabled the mechanical team to rapidly produce a functional prototype by leveraging core engineering principles and prior fabrication experience.
Fuselage Middle	The middle section of the fuselage responsible for holding most of the avionics	RCPlaneSkeletonMiddle1 • First draft of the front bringing the 3d sketch to reality RCPlaneSkeletonMiddle2 • Shortened the geometry as initial design was too large to 3d print in one piece RCPlaneSkeletonMiddle3 • Added cross members RCPlaneTop1 • Design switch from 3d modelling and printing to 2d CAD and laser cutting. Implemented a live hinge to create a seamless, strong one-piece fuselage RCPlaneTop2 • Scrapped the live hinge idea and shortened to original 3d sketch geometry RCPlaneTop3	

		• Added bolt holes and area for servo wires from wings to connect to microcontroller in fuselage. This area also acts as cooling for the battery and esc and lets antenna reach exterior of plane	
Fuselage Tail	The tail section of the fuselage	RCPlaneSkeletonTail • First draft of the front bringing the 3d sketch to reality RCPlaneSkeletonTailFront • Shortened the geometry as initial design was too large to 3d print in one piece RCPlaneSkeletonTailRear1 • Shortened the geometry as initial design was too large to 3d print in one piece RCPlaneSkeletonTailRear2 • Improved Geometry	
Wings and Horizontal Stabilizer	Wings, horizontal stabilizer and control surfaces that create the lift for the plane		Definitive conclusions remain pending successful flight testing. While the integrated systems currently function as intended, dynamic flight loads present risks to the wing and stabilizer connections. To mitigate potential failure, epoxy joints and bolt interfaces have been reinforced, ensuring structural integrity can withstand the aerodynamic stresses of the maiden flight.

Landing Gear	The landing gear responsible for taxiing on runway		The initial tricycle landing gear was replaced with a tail-dragger configuration to streamline assembly under project time constraints. For future refinement, integrating foam dampening on the rear caster wheel would significantly reduce motor-induced vibrations, improving ground handling and ensuring a more stable taxi during the takeoff roll.
Servos and Control Linkages	Responsible for moving the control surfaces to maneuver during flight		High-gauge steel pushrods with Z-bend attachments were utilized to translate servo rotation into surface deflection. While this interface was stress-tested to minimize "slop" and prevent aerodynamic flutter from the 4S prop-wash, the linkages remain high-maintenance components. Due to their precision-dependent nature, constant inspection and manual tensioning are required to maintain flight safety and prevent mechanical drift.

Electrical Systems

A. Main Motor Power System Specifications:
   1. Power Input and Output: The motor's maximum ratings are 650W at 14.8V (4S battery). This motor is then paired with a 60A ESC (maximum of 6S) and a 2200mAh 4S battery which is capable of 50C bursts. The ESC and battery were selected to give enough headroom for any power surges from the motor, acting as insurance for our first plane.
   2. Propeller and Thrust Estimation: Our original maximum weight was decided to be 1500g, meaning that all parts were selected with that in mind. A 9x6 propeller was the appropriate choice for our motor and weight class. With our motor's unloaded maximum RPM being 21,000 RPM, it is calculated that at 75% efficiency (13,200 RPM) the static thrust with our propeller will be 1600g.

B. Math
   1. Minimum Flight Time:

$$Capacity\ (C): 2200mAh, Maximum\ Current\ Draw\ (I): 43A$$

$$Time\ (t) = \frac{C}{I} = \frac{2200mAh}{43A} * \frac{1Ah}{1000mAh} = 0.05h = 3\ minutes$$

2. Static Thrust:
    a. Prop Diameter: *9 Inch*
    b. Prop Pitch: *6 Inch*
    c. Prop Static Rpm: *13,200 RPM*
    d. Supply Voltage: *14.8V*
    e. Supply Current: *43A*

   (Results):
    f. Estimated Static Thrust: *1638g*
    g. Supplied Power: *636W*
    h. Static Efficiency: *74.5%*

   An online calculator was used, Link: https://rcplanes.online/calc_thrust.htm.

The design and behaviour of the controller and plane circuits are essentially the same from the first and last versions (not including the prototypes in between). The communications reliability was the biggest upgrade between the two versions with it being optimized for longer ranges. For the sake of time, features from the first version were left unprogrammed (multicore, display, gyroscope). If a part is mentioned (IMU, display) then assume it is the same part from the previous version if not specified.

Controller

V1 Associated Files: RC-Plane_controller.ino	The first version was designed to only test the functionality of the antenna between the controller and a mock plane. The controller would transmit the joystick and throttle positions and receive the IMU data from the plane with the antenna. The controller would then draw the positions of the analog controls, the IMU data, and the connection status with the plane on the OLED display. The components in the circuit consisted of the following: • WaveShare RP2350-Zero (1x) • NRF24L01 PA + LNA Antenna Module (1x) • Joystick Module (1x) • Slide Potentiometer (1x) • SSD1306 128x64 OLED Display (1x) • 10nF Ceramic Capacitors (3x)
V2 (Final) Associated Files: controller.ino Controller_Final.kicad_pro	The second version kept all the original components with the new additions being the battery power + charging system and a rotary encoder that was never used in the program. The added components are as follows: • TP4056 Single Cell Battery Charger (1x) • XL6009 Boost Converter (1x) • Lithium Cell (unknown capacity) (1x) • Toggle switch (1x) • EC11 Rotary Encoder (1x)

	• 100uF Electrolytic Capacitor (1x) • Additional ceramic caps and resistors While this version had all these components, the display and rotary encoders were never given functions due to program issues. The controller will send the PWM signals or the motor and servos, along with an RTT timer for determining the input latency. The controller now also utilizes the microcontroller's on-board LED for displaying the connection status.

Final Controller Behaviour Description:

On startup, the controller will turn on the on-board LED and begin the antenna and serial communication. The serial communication is only for debugging purposes and displays the values being written to the motors, RTT, link status, and program runtime. Once initialized, the controller will attempt to connect to the plane until a link is established, when the link is lost or unavailable there is a 100ms timeout before it is considered lost which it will then change the colour of the status LED to red. While the link is established (the LED is green), the controller will read the analog inputs, convert the inputs into the required microsecond values for the motors, and transmit those values at 100Hz to the plane, which is double the frequency of the motor PWM signals. When converting the analog inputs to PWM signals, the controller has hardcoded trim and signal range values that it uses. The controller will always be looping in one of these two states and always listening for any transmissions from the plane.

The screen and rotary encoder were not given functionality in the final version as the coding required was too complex in the time left after this version was assembled. The screen was supposed to display real time data on the status of the plane and have a menu for modifying the trim and range values of each of the motors. The rotary encoder was meant to be the input for that, with the rotary encoder's switch acting as a selection button.

Plane

V1 (Mock Plane) Associated Files: RC-Plane_plane.ino	The first version of the plane, the "mock plane", was meant to only act as a plane by receiving the analog control data and transmitting the IMU data. It would draw a mirror of the controller's display to show that it was properly receiving the data. The circuit consisted of the following: • Waveshare RP2350-Zero (1x) • NRF24L01 PA + LNA Antenna Module (1x) • SSD1306 128x64 OLED Display (1x) • MPU6050 Inertial Measurement Unit (1x) This was built to test the functionality of the antenna.
V2 (Final) Associated Files:	The second version was fully rebuilt as a soldered control board with pinouts for each module to reduce its size and weight. The board uses a mix of male and female pin headers so that all the on-board modules can be removed and all the external

plane.ino Plane_Final.kicad_pro	modules can be disconnected from the board. The control board consists of just the microcontroller and the IMU, with the microcontroller's on-board LED displaying the connection status. The antenna is detached so that it can be optimally positioned, the antenna cable also has grounded cable shielding. The control board has five PWM channels, one for the motor and 4 for servos (we only used 3). The motor channel connects to the ESC which powers the board from the BEC (5v) and supplies power to the servos. The control board circuit consists of the following components: • Waveshare RP2350-Zero (1x) • MPU6050 Inertial Measurement Unit (1x) The antenna is a detached module with the following: • NRF24L01 PA + LNA Antenna Module (1x) • 10nF Ceramic Capacitors (1x) • 100uF Electrolytic Capacitor (1x) The rest of the plane consists of: • 1450Kv Brushless Motor (1x) • 60A ESC (1x) • 2200mAh 4S 14.8V Lithium Battery (1x) • MG90s Servos (3x) The antenna and electromechanical components are adhered or mounted to the plane.

Final Plane Behaviour Description:

On startup the plane will turn on the on-board LED, begin the serial debug output, and initialize the antenna. After initializing, the plane will then wait for the link with the controller to be established. Once the connection is established (LED set to green), the plane will begin receiving the motor values and will continuously write them to the motors while the link is established. Additionally, the plane will transmit IMU data which was changed to being just an 8-bit counter. When the connection is lost (LED set to red), the plane will cut the motor and set all the servos to their default position (currently flat), the plane will stay in this state until the link is reestablished. This is the whole loop.

During testing it was found that the IMU was blocking the loop, leading to the plane being unable to establish a consistent connection with the controller. Because of this, the IMU was taken out of the loop so that it no longer is blocking, and because there was already a set packet structure the "IMU data" was replaced with a looping 8-bit counter so that minimal changes had to be made. If the IMU's functionality would be restored, the safest way it would be done is running it on the second core with a transmission memory buffer, this wasn't done for the same reason as the controller's display.

Conclusion

The mechanical phase of this project successfully transitioned a conceptual design into a functional, 1250g prototype aircraft. By pivoting from 3D printing to laser-cut plywood, the team achieved the necessary torsional rigidity required to support high-torque motor operations while maintaining a manageable fabrication timeline. Mathematical validation through the lift equation ensured a 55% lift surplus, providing a substantial safety margin for upcoming flight trials.

On the electronics side, the first and second versions were successful, with some features unfortunately needing to be cut from the final version. While running and operational, the code would need to be refined and optimized in a future version along with multicore support for handling peripherals at both ends. At the ends of the final version, focusing on getting the plane operational was the best option. The remaining uncertainty is the antenna and it’s link stability as while it is rated for up to 1100m, during antenna testing it was found that it may only be effective in low noise environments or when the transmitter and receiver are aligned.

While the airframe and propulsion systems are fully integrated, final conclusions regarding flight performance remain pending the results of the maiden test flight. Future iterations will focus on refining the control linkages to reduce manual maintenance and implementing vibration dampening in the landing gear. Combined with the software and avionics integration, the current progress demonstrates a viable platform for stable, remote-controlled flight.

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